

Name: _____

Date: _____

TRIGONOMETRY, (EXACT TRIGONOMETRIC VALUES)

LO: I can recall and use exact trigonometric values for 0, 30, 45, 60 and 90 degrees.

Exact trig values

The **exact trigonometric values** for common angles (0° , 30° , 45° , 60° , 90°) can be expressed as **fractions and surds** rather than decimals.

Key values: $\sin 30^\circ = \frac{1}{2}$, $\cos 60^\circ = \frac{1}{2}$, $\tan 45^\circ = 1$, $\sin 45^\circ = \frac{\sqrt{2}}{2}$.

Values to memorise

	$\sin 30^\circ = \frac{1}{2}$	from 30-60-90 triangle
	$\cos 45^\circ = \frac{\sqrt{2}}{2}$	from 45-45-90 triangle
	$\tan 60^\circ = \sqrt{3}$	$\sin 60 / \cos 60$
	$\sin 45^\circ = \frac{\sqrt{2}}{2}$	$\sin 45 = \frac{\sqrt{2}}{2}$
	$\cos 90^\circ = 0$	$\cos 90 = 0$

Key idea

Learn the exact values from the 30-60-90 and 45-45-90 triangles.

$$\sin 60^\circ = \frac{\sqrt{3}}{2} \quad | \quad \cos 30^\circ = \frac{\sqrt{3}}{2}$$

sin and cos are complementary: $\sin x = \cos(90 - x)$

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - sin values

State the exact value of $\sin 60^\circ$.

From the 30-60-90 triangle
Opposite = $\sqrt{3}$, hypotenuse = 2
 $\sin 60^\circ = \frac{\sqrt{3}}{2}$

$$\sin 60^\circ = \frac{\sqrt{3}}{2}$$

The 30-60-90 triangle has sides 1, $\sqrt{3}$, 2.

Example 2 - cos values

Find the exact value of $\cos 30^\circ$.

From the 30-60-90 triangle
Adjacent = $\sqrt{3}$, hypotenuse = 2
 $\cos 30^\circ = \frac{\sqrt{3}}{2}$

$$\cos 30^\circ = \frac{\sqrt{3}}{2}$$

$\cos 30 = \sin 60$ (complementary angles).

Example 3 - tan values

Find the exact value of $\tan 45^\circ$.

From the 45-45-90 triangle
Opposite = adjacent = 1
 $\tan 45^\circ = \frac{1}{1} = 1$

$$\tan 45^\circ = 1$$

In a 45-45-90 triangle, the two legs are equal.

Example 4 - using exact values

Find the hypotenuse when the opposite is 8 and angle is 30° .

$\sin 30^\circ = \text{opp/hyp} = 8/\text{hyp}$
 $\frac{1}{2} = 8/\text{hyp}$
 $\text{hyp} = 16$

$$\text{Hypotenuse} = 16$$

Substitute the exact value of $\sin 30^\circ = \frac{1}{2}$.

YOUR TURN – PRACTICE (deliberate practice)

1. Recall sin values

State the exact value.

- a) $\sin 0^\circ$
- b) $\sin 30^\circ$
- c) $\sin 45^\circ$
- d) $\sin 90^\circ$

2. Recall cos values

State the exact value.

- a) $\cos 0^\circ$
- b) $\cos 30^\circ$
- c) $\cos 60^\circ$
- d) $\cos 90^\circ$

3. Recall tan values

State the exact value.

- a) $\tan 0^\circ$
- b) $\tan 30^\circ$
- c) $\tan 45^\circ$
- d) $\tan 60^\circ$

4. Finding lengths

Use exact values to find the unknown.

- a) $\sin 30^\circ = x/10$. Find x .
- b) $\cos 60^\circ = 6/h$. Find h .
- c) $\tan 45^\circ = y/7$. Find y .
- d) $\sin 60^\circ = 9/h$. Find h .

5. Mixed exact values

Calculate exactly.

- a) $\sin 30^\circ + \cos 60^\circ$
- b) $\sin^2 45^\circ + \cos^2 45^\circ$
- c) $2 \sin 30^\circ \cos 30^\circ$
- d) $\tan 60^\circ - \tan 30^\circ$
- e) $\sin 60^\circ \div \cos 60^\circ$

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) $\sin 30^\circ = \cos 60^\circ = \frac{1}{2}$ ☐

Correct answer: _____

Reason: _____

b) $\sin 45^\circ = \frac{1}{2}$ ☐

Correct answer: _____

Reason: _____

c) $\tan 45^\circ = 1$ ☐

Correct answer: _____

Reason: _____

d) $\sin 90^\circ = 0$ ☐

Correct answer: _____

Reason: _____

e) $\sin^2 x + \cos^2 x = 1$ for any angle x ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) An equilateral triangle has side length **10 cm**. Using exact values, find the height and area.

Expression: _____

Simplified: _____

b) Show that $\sin^2 60^\circ + \cos^2 60^\circ = 1$. Find $\sin 60^\circ \cos 30^\circ + \sin 30^\circ \cos 60^\circ$.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

